# QUANTUMFLOOR

*Computational Research Report*

Job ID: 433d5fae

## 1. Executive Summary

This report details the execution of the QuantumFloor diagnostic pipeline on dataset 'label'. The system evaluated multiple classical baselines against quantum algorithms to determine the optimal strategy.

## 2. Dataset Intelligence

| | |
|---|---|
| **Modality:** | TABULAR |
| **Total Rows:** | 80 |
| **Unique Rows:** | 80 |
| **Duplicate Rows:** | 0 |
| **Total Columns:** | 5 |
| **Missing Values:** | 0.00% |
| **Leakage Risk Detected:** | 0 |

## 3. Data Preparation

| | |
|---|---|
| **Train/Test Split:** | 80/20 Stratified |
| **Preprocessing:** | Pipeline (Imputation + Scaling) |
| **Leakage Protection:** | Enforced |

## 4. Computational Plan

| | |
|---|---|
| **Original Features:** | 5 |
| **Quantum Embedding Dimension:** | 2 |
| **Qubit Budget:** | 2 |
| **Classical Training Size:** | 64 |
| **Quantum Subset Size:** | 64 |

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## 5. Quantum Execution

**Execution Backend:** SIMULATED EXECUTION

**Ansatz:** AngleEmbedding + IsingZZ

**Qubits:** 2

**Circuit Depth:** 2

**Total Operations:** 2

## 6. Classical vs Quantum Baselines

| Model | Type | AUROC | F1 | Spec@95 | Notes |
|---|---|---|---|---|---|
| full_feature_classical | Classical | 0.5078 | 0.5000 | 0.0000 | |
| logreg | Classical | 0.3281 | 0.5263 | 0.0000 | |
| svm_rbf | Classical | 0.3750 | 0.3077 | 0.0000 | |
| random_forest | Classical | 0.6875 | 0.6667 | 0.5000 | |
| qsvm | Quantum | 0.5625 | 0.4286 | 0.3750 | |
| pqk | Quantum | 0.7500 | 0.6667 | 0.6250 | |
| vqc | Quantum | 0.3438 | 0.6087 | 0.0000 | |

## 7. Comparison & Explainability

**Best Classical AUROC:** 0.6875

**Best Quantum AUROC:** 0.7500

**Delta:** 0.0625

**Explainability Status:** available

**PC1:** 0.1468

**PC2:** 0.1290

## 8. Final Verdict

**Best Overall Model:** pqk

**Quantum Advantage:** Yes